

DH Wrap Up

EGR 445/545

GVSU

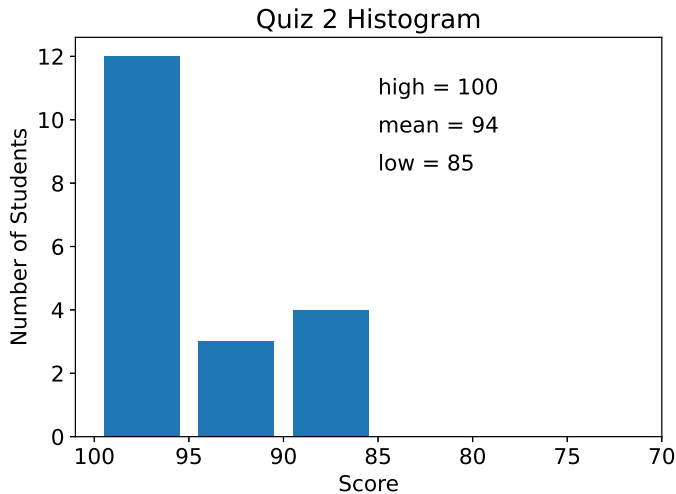
Plan for Today

- quiz 2 solution
- white vs. yellow Fanuc DH discussion
- PUMA DH discussion
- project 1 demos

Reminder

- Exam 1 Monday, 6/14
 - approx. 60 minutes
 - chapters 2 and 3
 - 1-2 conceptual questions
 - 3-4 questions with Python available
 - must start with “official blank” notebook
 - you may use the Python ‘help’ function
 - you may not use any other resources
 - 1 page of hand written notes
 - 8.5” by 11” US letter paper

Quiz 2 Histogram



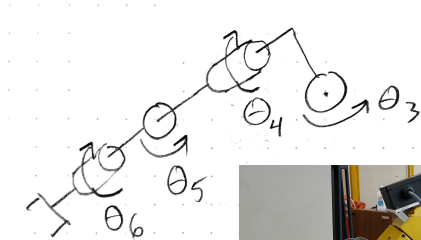
White vs. Yellow Fanuc Discussion

Yellow Fanuc Schematic

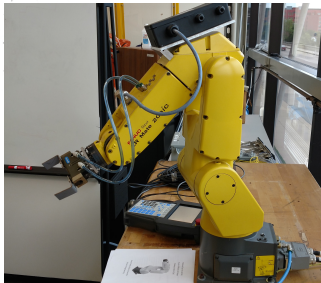
White Fanuc Schematic

- what is different?

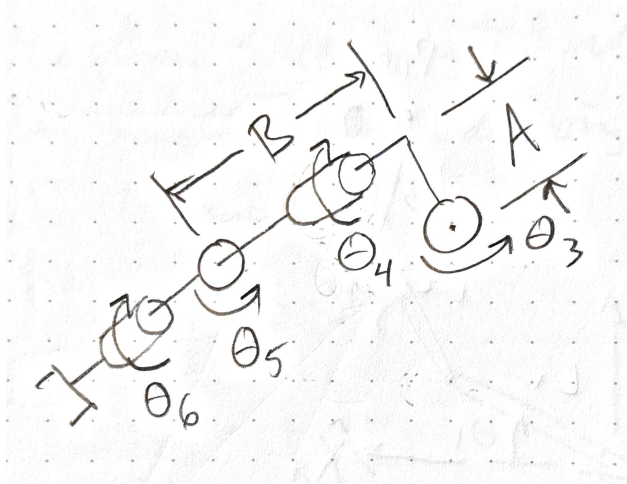
Yellow Fanuc Forearm/Wrist



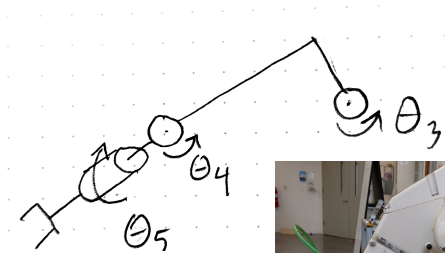
- where is O_3 ?
- do we have any flexibility in the location of O_3 ?



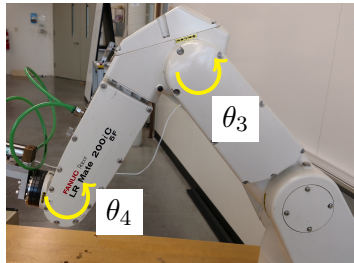
Yellow Fanuc: What do we call these two parameters?



White Fanuc Forearm/Wrist

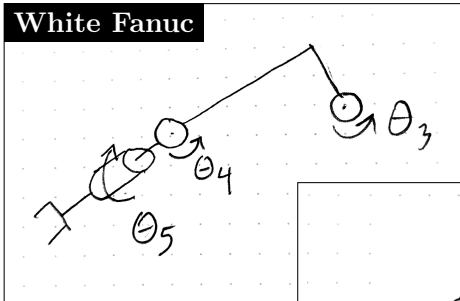


- only 5 joints
- what is different about $\hat{Z}_3$ and $\hat{Z}_4$?
- where is O_3 ?
- what are a_3 and d_4 ?

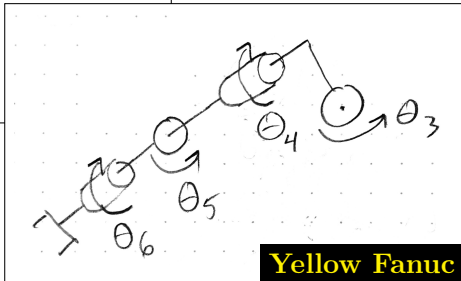


Comparing the White and Yellow Fanucs

White Fanuc



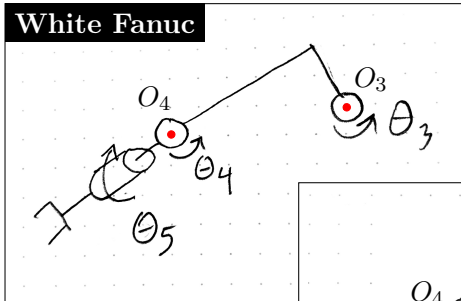
- where is O_3 for both robots?
- do we have any flexibility?



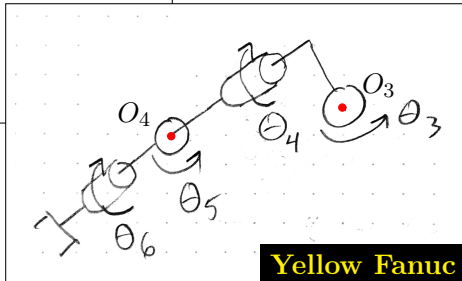
Yellow Fanuc

Comparing the White and Yellow Fanucs

White Fanuc



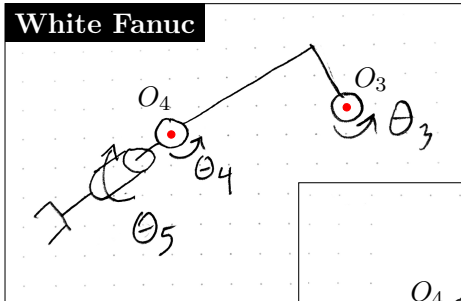
- where does $\hat{X}_3$ point for both robots?



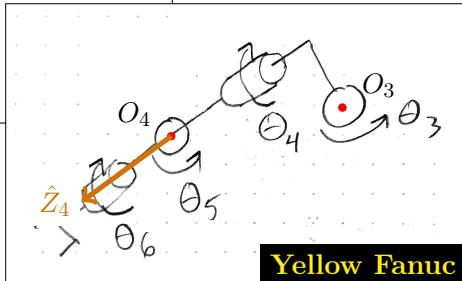
Yellow Fanuc

Comparing the White and Yellow Fanucs

White Fanuc



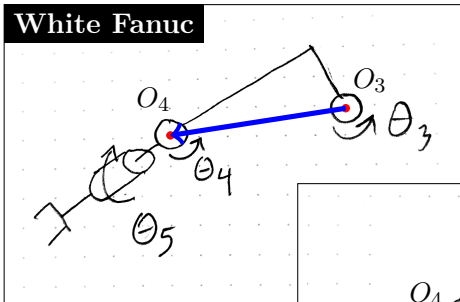
- where does $\hat{X}_3$ point for both robots?



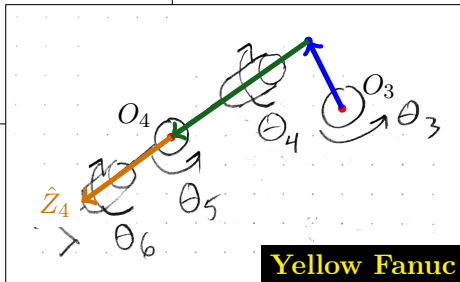
Yellow Fanuc

Comparing the White and Yellow Fanucs

White Fanuc



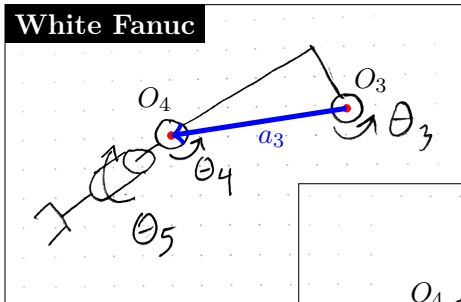
- what do we call these distances?



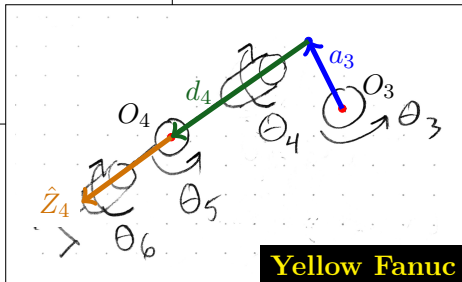
Yellow Fanuc

Comparing the White and Yellow Fanucs

White Fanuc



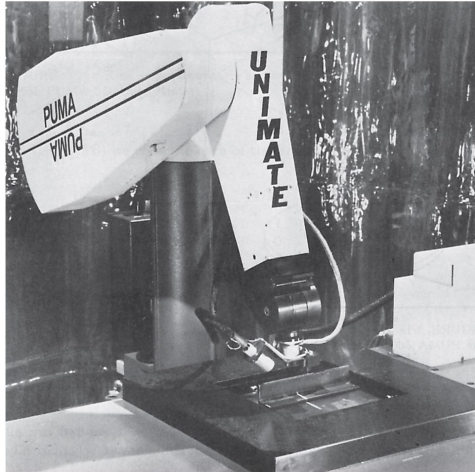
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Yellow Fanuc

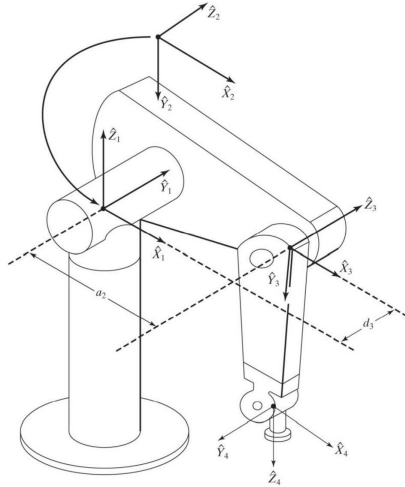
PUMA 560

- section 3.7, starting on page 83



Courtesy of Unimation Incorporated, Shelter Rock Lane, Danbury, Conn.

PUMA Main Schematic (Fig. 3.18)

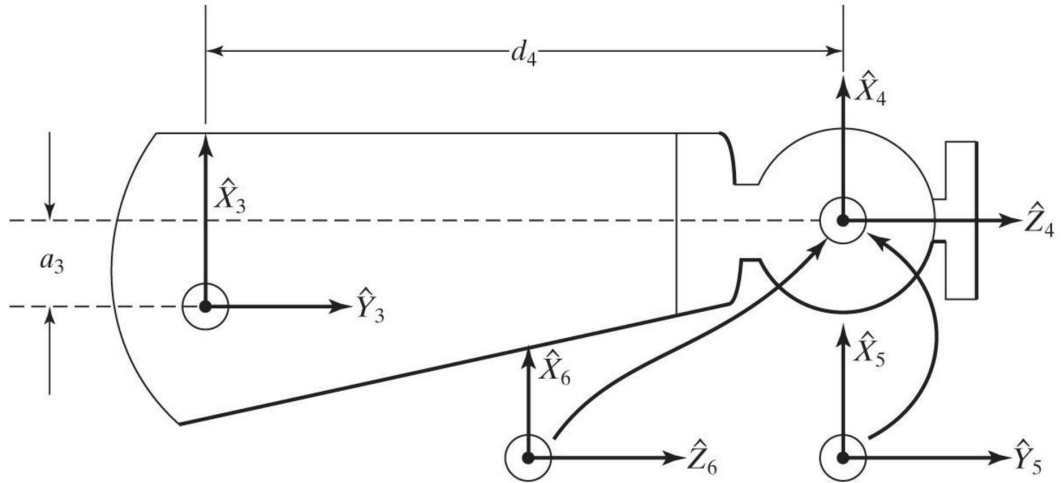


YouTube Animation

From Ohio University:

https://www.youtube.com/watch?v=ArzP7rh4_9Q

PUMA Forearm (Fig. 3.19)



PUMA Wrist (Fig. 3.20)

